

AI-Driven Fraud Detection System

Rulebook, Risk Scoring & Investigation Guide

1. INTRODUCTION

This document defines the fraud detection logic used in the system. The system combines:

- Rule-based detection (deterministic)
- Probabilistic fraud scoring
- Behavioral analysis
- AI-assisted investigation

The purpose is to:

- Detect suspicious transactions in real time
 - Assign a fraud risk score
 - Provide explainable reasoning for decisions
 - Assist fraud analysts in investigation
-

2. SYSTEM OVERVIEW

Each transaction is evaluated using multiple layers:

Transaction → Rule Engine → Rule Signals → Fraud Score → Final Decision

Final Output:

- **VALID** → Safe transaction
 - **ALERT** → Suspicious, needs monitoring
 - **FRAUD** → High-risk, should be blocked
-

3. FRAUD SCORING MODEL

Each rule contributes a **risk signal (0 to 1)**.

Scoring Formula:

Fraud Score = $1 - \prod(1 - \text{RuleScore})$

Interpretation:

Score Range	Decision
0.00 – 0.49	VALID
0.50 – 0.84	ALERT
0.85 – 1.00	FRAUD

4. RULE DEFINITIONS

4.1 BLACKLISTED ACCOUNT RULE

Description:

Checks whether the sender or receiver account is present in the blacklist database.

Condition:

- If sender OR receiver is blacklisted

Output:

- FRAUD

Risk Score:

- 1.0 (Critical)

Reason:

Blacklisted accounts are confirmed fraud entities or previously involved in suspicious activity.

Analyst Action:

- Immediately block transaction
 - Freeze account
 - Trigger investigation
-

4.2 CREDIT LIMIT EXCEEDED RULE

Description:

Validates whether a credit transaction exceeds the defined account limit.

Condition:

- Transaction type = CREDIT
- Amount > account transaction limit

Output:

- FRAUD

Risk Score:

- 0.9

Reason:

Exceeding credit limits indicates misuse, fraud, or system abuse.

Analyst Action:

- Verify transaction legitimacy
 - Check user authorization
 - Review historical credit usage
-

4.3 HIGH VALUE TRANSACTION RULE

Description:

Flags transactions exceeding ₹10,00,000.

Condition:

- Amount > 1,000,000

Output:

- FRAUD

Risk Score:

- 0.8

Reason:

High-value transactions are often used for fraud laundering or rapid fund movement.

Analyst Action:

- Check transaction source
 - Verify user identity
 - Inspect destination account
-

4.4 INTERNATIONAL TRANSACTION RULE

Description:

Detects high-value foreign currency transactions.

Condition:

- Currency ≠ INR
- Amount > 100,000

Output:

- ALERT

Risk Score:

- 0.7

Reason:

Cross-border transactions carry higher fraud risk due to jurisdiction differences.

Analyst Action:

- Verify destination country
 - Check transaction frequency
 - Look for unusual patterns
-

4.5 NIGHT TRANSACTION RULE

Description:

Detects transactions occurring during unusual hours.

Condition:

- Time between **12:00 AM – 4:00 AM**

Output:

- **ALERT**

Risk Score:

- **0.4**

Reason:

Fraudulent activities often occur during low monitoring periods.

Analyst Action:

- Check user activity patterns
 - Validate login sessions
 - Review device/location
-

4.6 VELOCITY CHECK RULE

Description:

Detects multiple rapid transactions from the same account.

Condition:

- Multiple transactions within **5 seconds**

Output:

- FRAUD

Risk Score:

- 0.9

Reason:

Indicates automated scripts or bot activity.

Analyst Action:

- Block account temporarily
 - Investigate transaction burst
 - Check API misuse
-

4.7 RAPID CROSS-BORDER RULE

Description:

Detects repeated foreign transactions from the same account.

Condition:

- Multiple non-INR transactions in short time

Output:

- FRAUD

Risk Score:

- 0.85

Reason:

Indicates potential money laundering or cross-border fraud patterns.

Analyst Action:

- Monitor transaction trail
- Check international beneficiaries
- Notify compliance team

4.8 REPEATED TRANSACTION RULE

Description:

Detects identical transactions (same sender, receiver, amount).

Condition:

- Same transaction repeated within short duration

Output:

- **ALERT**

Risk Score:

- **0.5**

Reason:

Could indicate testing, duplication, or fraud automation.

Analyst Action:

- Check transaction logs
 - Verify user intent
-

4.9 ROUND AMOUNT RULE

Description:

Flags transactions with round figures.

Condition:

- $\text{Amount} \% 100000 == 0$
- $\text{Amount} \leq 1,000,000$

Output:

- **ALERT**

Risk Score:

- 0.3

Reason:

Fraudsters often use round figures for testing systems.

Analyst Action:

- Cross-check with transaction history
 - Look for repeated patterns
-

4.10 SUSPICIOUS MERCHANT RULE

Description:

Checks if the receiver is a flagged merchant.

Condition:

- Receiver present in suspicious merchant database

Output:

- ALERT

Risk Score:

- 0.6

Reason:

Known suspicious merchants are frequently linked to fraud.

Analyst Action:

- Investigate merchant profile
 - Check prior fraud cases
-

4.11 WEEKEND TRANSACTION RULE

Description:

Flags high-value weekend transactions.

Condition:

- Day = Saturday or Sunday
- Amount > 500,000

Output:

- **ALERT**

Risk Score:

- **0.5**

Reason:

Weekend transactions may bypass monitoring or involve delayed verification.

Analyst Action:

- Validate transaction timing
 - Check user availability
-

5. COMBINED FRAUD ANALYSIS

Multiple rules can trigger simultaneously.

Example:

Transaction:

Amount: 1,200,000

Time: 2 AM

Currency: USD

Triggered Rules:

- High Value → 0.8
- Night → 0.4
- International → 0.7

Fraud Score:

Score = $1 - (1-0.8)*(1-0.4)*(1-0.7)$
≈ 0.94 → FRAUD

6. FRAUD EXPLANATION FORMAT (USED BY AI)

The system generates explanations like:

Transaction flagged as FRAUD because:

- High-value transaction exceeding ₹10L
- Occurred during suspicious night hours
- Involved international transfer

Fraud Score: 0.94

7. INVESTIGATION WORKFLOW

When a transaction is flagged:

Step 1:

Check triggered rules

Step 2:

Review fraud score

Step 3:

Analyze account history

Step 4:

Validate user identity

Step 5:

Take action:

- Allow
- Monitor
- Block

8. AI ASSISTANT USAGE

The AI assistant uses this knowledge base to:

- Explain why a transaction was flagged
 - Suggest investigation steps
 - Interpret fraud patterns
 - Combine DB + rule knowledge
-

9. SAMPLE QUERIES

Query:

Why was my transaction flagged?

Response:

Your transaction was flagged due to:

- High transaction amount
- Suspicious timing
- Cross-border activity

This matches known fraud patterns.

10. CONCLUSION

This system combines:

- Rule-based detection
- Probabilistic scoring
- Behavioral monitoring
- AI-powered explanation